METRIC

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10 November 1997

SUPERSEDING
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PERFORMANCE SPECIFICATION

INHIBITOR, CORROSION/LUBRICITY IMPROVER, FUEL SOLUBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of fuel-soluble corrosion inhibitor/lubricity improver additive for use in aviation turbine fuel, motor gasoline, diesel fuel, and related petroleum products.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASC/ENSI, Bldg 560, 2530 Loop Road West, Wright-Patterson AFB OH 45433-7101, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 6850

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2.2 Government documents

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-52557	Fuel Oil, Diesel; for Posts, Camps and Stations
TT-S-735	Standard Test Fluids, Hydrocarbon

DEPARTMENT OF DEFENSE

MIL-G-3056	Gasoline, Automotive, Combat
MIL-PRF-5624	Turbine Fuel, Aviation, Grades JP-4, JP-5, and JP-5/JP-8 ST
MIL-C-7024	Calibrating Fluids, Aircraft Fuel System Components
MIL-PRF-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-T-25524	Turbine Fuel, Aviation, Thermally Stable
MIL-F-46162	Fuel, Diesel, Referee Grade
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35
MIL-I-85470	Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745

STANDARDS

FEDERAL

FED-STD-791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia PA 19111-5094, [215] 697-2179.)

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2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS, INC. (ASTM)

ASTM A108	Standard Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality (DoD adopted)
ASTM D56	Standard Test Method for Flash Point by Tag Closed Tester (DoD adopted)
ASTM D93	Standard Test Methods for Flash-Point by Pensky-Martens Closed Cup Tester (DoD adopted)
ASTM D97	Standard Test Method for Pour Point of Petroleum Products (DoD adopted)
ASTM D445	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (the Calculation of Dynamic Viscosity) (DoD adopted)
ASTM D482	Standard Test Method for Ash From Petroleum Products (DoD adopted)
ASTM D664	Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration (DoD adopted)
ASTM D665	Standard Test Method for Rust-Preventing Characteristics of Inhibited Mineral Oil in the Presence of Water (DoD adopted)
ASTM D1298	Standard Practice for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method (DoD adopted)
ASTM D2274	Standard Test Method for Oxidation Stability of Distillate Fuel Oil (Accelerated Method) (DoD adopted)
ASTM D2624	Standard Test Methods for Electrical Conductivity of Aviation and Distillate Fuels
ASTM D3948	Standard Test Method for Determining Water Separation Characteristics of Aviation Turbine Fuels by Portable Separometer (DoD adopted)
ASTM D4052	Standard Test Method for Density and Relative Density of Liquids by Digital Density Meter (DoD adopted)
ASTM D4057	Standard Practice for Manual Sampling of Petroleum and Petroleum Products (DoD adopted)
ASTM D4308	Standard Test Method for Electrical Conductivity of Liquid Hydrocarbons by Precision Meter (DoD adopted)
ASTM D4814	Standard Specification for Automotive Spark-Ignition Engine Fuel
ASTM D5001	Standard Test Method for Measurement of Lubricity of Aviation Turbine Fuels by the Ball-on-Cylinder Lubricity Evaluator (BOCLE)
ASTM D5949	Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method)
ASTM D5950	Standard Test Method for Pour Point of Petroleum Products (Automatic Tilt Method)
ASTM D 5985	Standard Test Method for Pour Point of Petroleum Products (Rotational Method)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken PA 19428-2959; [610] 832-9500.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulation unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The corrosion inhibitor/lubricity improvers furnished under this specification shall be products that are authorized by the Qualifying Activity for listing on the applicable Qualified Products List before contract award (see 4.2 and 6.6).

3.1.1 Qualification requirements. The qualification requirements for the inhibitors are listed for each type of fuel. All approved inhibitors shall meet the requirements of 3.2 through 3.12, 3.16, and 3.17 to be qualified for use in fuels which conform to MIL-PRF-5624, MIL-C-7024, MIL-T-25524, and MIL-T-83133. To qualify for use in motor gasolines (MIL-G-3056 and ASTM D4814) and diesel fuels (A-A-52557), the inhibitor shall meet the requirements of 3.2 through 3.17. (See table I.)

3.2 Materials. The composition of the finished inhibitor is not limited but is subject to review by the Qualifying Activity to ensure service compatibility with previously qualified products.

3.2.1 Toxic products and formulations. The materials shall have no adverse effect on the health of personnel when used for their intended purpose. Questions pertinent to this effect shall be referred by the Procuring Activity to the appropriate departmental medical service, which will act as an advisor to the Procuring Activity (see 6.4).

3.3 Solubility. The maximum allowable concentration of inhibitor, as defined in 3.7, shall be readily and completely dissolvable in all fuels for which it is qualified. There shall be no precipitation, cloudiness, or other evidence of insolubility when tested as specified in 4.4.1.

3.4 Compatibility. The inhibitor shall be compatible with all inhibitors currently qualified under this specification, with all the static dissipater additive(s) allowed in MIL-T-83133, and with all the thermal stability improver additive(s) allowed in MIL-T-83133. There shall be no precipitation, cloudiness, or other evidence of noncompatibility when tested as specified in 4.4.2.

3.5 Relative effective concentration. The relative effective concentration shall be determined in accordance with 4.4.3 and shall be expressed in grams of finished inhibitor per cubic meter of fuel. The relative effective concentration shall not be less than 6 grams of inhibitor per cubic meter of fuel (6 g/m^3), not more than 36 g/m^3, and shall be approved at concentrations divisible by 3 (i.e.; 6, 9, 12, 15, ... 36 g/m^3). The relative effective concentration shall be identified by the Qualifying Activity and cited on QPL-25017.

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TABLE I. Testing.

Characteristic	Qualification Test	Conformance Test	Test Paragraph	Test Method Paragraph
Material Compatibility	Required		3.2	
Solubility in Fuel	Required		3.3	4.4.1
Compatibility with other Fuel Additives	Required		3.4	4.4.2
Rust Test	Required		3.5	4.4.3.1
Lubricity Test	Required	Required	3.6	4.4.4.1, ASTM D5001
Water Separation	Required	Required	3.7	4.4.5.1, ASTM D3948
Electrical Conductivity	Required		3.7	4.4.2.2, ASTM D2624 or ASTM D4308
Ash Content	Required	Required	3.8	4.4.6, ASTM D482
Pour Point	Required	Required	3.9	4.4.7, ASTM D97 or ASTM D5949 or ASTM D5950 or ASTM D5985
Engine Operation	Required		3.10	4.4.8
Fuel Specification Properties	Required		3.11	4.4.9
Storage Stability	Required		3.12	4.4.10
Induction System Deposit Test	Required for Category 1 Qualification		3.13	4.4.11, Method 500 of FED-STD-791
Emulsification Tendency	Required for Category 1 Qualification		3.14	4.4.12, Method 550 of FED-STD-791
Accelerated Stability	Required for Category 1 Qualification		3.15	4.4.13, ASTM D2274
Density	Required	Required	3.16	4.4.14, ASTM D1298 or pycnometer or ASTM D4052
Viscosity	Required	Required	3.16	4.4.14, ASTM D445
Flash Point	Required	Required	3.16	4.4.14, ASTM D56 or ASTM D93
Neutralization Number	Required	Required	3.16	4.4.14, ASTM D664
pH	Required		3.16	4.4.14, ASTM D664
Metallic Constituent	Required		3.16	4.4.14
Workmanship	Required		3.17	

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3.6 Minimum effective concentration. The minimum effective concentration shall be the larger of the following:

a. One and one-half (1.5) times the relative effective concentration, as defined in 3.5, rounded to the next whole number of grams per cubic meter (i.e.; 9, 14, 18, 23, ... 54 g/m^3).

b. The amount of inhibitor (in whole numbers of grams per cubic meter; e.g.; 9, 10, 11, 12, ... g/m^3) that gives a wear scar diameter of 0.65 mm or less when tested for lubricity using the Ball-on-Cylinder Lubricity Evaluator (BOCLE), as specified in 4.4.4.1.

The minimum effective concentration shall be identified by the Qualifying Activity and cited on QPL-25017.

3.7 Maximum allowable concentration. The maximum allowable concentration shall be the lowest of the following (all expressed in grams of inhibitor per cubic meter of fuel):

a. Fifty-four grams of inhibitor per cubic meter of fuel (54 g/m^3)

b. Four times the relative effective concentration, as defined in 3.5

c. The highest concentration that results in a Micro-Separometer Rating (MSEP) of 70 or higher when determined in accordance of 4.4.5.1.

d. The highest concentration that results in a less-than-40-percent change in electrical conductivity with fuel that contains static dissipater additive (see 4.4.2.2).

The maximum allowable concentration shall be identified by the Qualifying Activity and cited on QPL-25017.

3.8 Ash content. The ash content of the inhibitor shall not exceed 0.10 percent when determined in accordance with 4.4.6.

3.9 Pour point. The maximum allowable pour point of the finished inhibitor shall be –18°C when determined as specified in 4.4.7.

3.10 Aircraft turbine engine operation. Grade JP-8 (MIL-T-83133) or JP-5 (MIL-PRF-5624) fuel that contains twice the maximum allowable concentration (see 3.7) of the inhibitor shall be tested in accordance with 4.4.8 to determine its acceptability for turbine engine use. Engine operation shall not be adversely affected and the post-test condition of the engine shall indicate no excessive deposits, water, corrosion, or other adulteration, attributable to the inhibitor.

3.11 Specification requirements. A blend of the inhibitor at its maximum allowable concentration in a representative fuel shall meet all of the requirements of each applicable specification when tested in accordance with 4.4.9. For example, to be qualified for use in a motor gasoline, a gasoline that conforms to MIL-G-3056 shall continue to meet all applicable requirements of MIL-G-3056 after the maximum allowable concentration of the inhibitor is added.

3.12 Storage stability. After stored for 12 months in accordance with 4.4.10, the inhibitor shall show no precipitation, layering, or other evidence of gross separation or degradation. Inhibitor that represents the top-half of the stored sample shall meet all requirements of this specification except 3.10.

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3.13 Induction system deposit. Inhibitor for use in motor gasolines which conform to ASTM D4814 and MIL-G-3056 shall pass the induction system deposit test performed in accordance with 4.4.11.

3.14 Emulsification tendency. To obtain approval for use in motor gasolines which conform to ASTM D4814 and MIL-G-3056 and diesel fuels which conform to A-A-52557, the inhibitor shall pass the emulsification tendency test performed in accordance with 4.4.12.

3.15 Accelerated stability. The inhibitor shall pass the accelerated stability test performed in accordance with 4.4.13 before approved for use in diesel fuels which conform to A-A-52557.

3.16 Identification qualification data. The following properties of the finished inhibitor shall be determined but not limited during qualification: density at 15°C, viscosity at 40°C, flash point, neutralization number, pH, and type of metallic constituent, if present (see 4.4.14). The permissible production variation of individual properties shall be established at the time of qualification by mutual agreement between the manufacturer and the Qualifying Activity. Individual batches of inhibitor subsequently subjected to qualification conformance inspections shall conform to the established range of properties. The ranges shall not adversely affect any of the inhibitor performance characteristics such as relative effective concentration and MSEP.

3.17 Workmanship. The finished product in bulk or container shall be uniform in appearance and visually free from grit, undissolved water, insoluble components, or other adulteration.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follow:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4.

4.2.1 Qualification sampling. Unless otherwise specified by the activity responsible for qualification, an initial 1-liter sample of finished inhibitor shall be submitted for evaluation by all the tests required by 3.1.1, with the exception of the storage stability (3.12) and aircraft turbine engine test (3.10), to an independent laboratory approved by the Qualifying Activity. If the product passes all the tests, an additional sample of finished inhibitor will be required for the storage stability and aircraft turbine engine tests. The additional samples shall be identified as required and forwarded to the Qualifying Activity (see 6.6).

4.2.2 Requalification. Requalification will be required in the event any change in composition or formulation, source of inhibitor or its ingredients, or manufacturing practice or site occurs.

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4.2.3 Retention of qualification. The retention of qualification of products approved for listing on the Qualified Products List (QPL) shall be accomplished by a periodic verification to determine continued compliance of a supplier's product with the requirements of this specification. The verification intervals shall not exceed two years. Unless otherwise specified by the activity responsible for the Qualified Products List, verification of qualification may be made by certification.

4.3 Conformance inspection. Quality conformance inspection of a bulk lot, prior to becoming a packaged lot, of inhibitor shall consist of tests for conformance requirements for ash content (3.8), pour point (3.9), and the property limits (3.16) shown on the Qualified Products List. In addition, the product shall be required to result in a MSEP of 70 or higher (average of three tests) when tested at the maximum allowable concentration in accordance with 4.4.5.1. The product shall also give a wear scar diameter of 0.65 mm or less when tested at the minimum effective concentration for lubricity using the Ball-on-Cylinder Lubricity Evaluator as specified in 4.4.4.1. (See table I.)

4.3.1 Inspection lot

4.3.1.1 Bulk lot. A bulk lot is defined as an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single, isolated container; manufactured through the same processing equipment; with no change in ingredient material.

4.3.1.2 Packaged lot. A packaged lot is defined as an indefinite number of 208-liter (55-gallon) drums or smaller unit packages of identical size and type, or an indefinite number of 2080-liter (550-gallon) or less returnable containers, such as totes, offered for acceptance, and filled with a homogenous mixture of material from a bulk lot.

4.3.2 Sampling. Each bulk or packaged lot of material shall be sampled for verification of product quality and compliance in accordance with ASTM D4057.

4.4 Methods of inspection

4.4.1 Solubility. The maximum allowable concentration of inhibitor shall be mixed with each of the following fuels. The fuel shall contain no other inhibitors. Immediately after mixing and at the end of 24 hours, the samples shall be visually inspected for precipitation, cloudiness, or other evidence of insolubility.

a. JP-4 fuel that conforms to MIL-PRF-5624 and contains the maximum allowable concentration of specified fuel system icing inhibitor

b. JP-5 fuel that conforms to MIL-PRF-5624 or JP-8 fuel that conforms to MIL-T-83133 and contains the maximum allowable concentration of specified fuel system icing inhibitor

c. A motor gasoline that conforms to MIL-G-3056 or ASTM D4814 (required only if inhibitor is to be qualified for use in motor gasolines)

d. A diesel fuel that conforms to A-A-52557 or MIL-F-46162 (required only if inhibitor is to be qualified for use in diesel fuel).

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4.4.2 Compatibility

4.4.2.1 Other inhibitors. Grade JP-8 fuel (MIL-T-83133) or grade JP-5 fuel (MIL-PRF-5624) that contains the maximum allowable concentration of the inhibitor under test and no other corrosion inhibitors shall be mixed in equal proportions with samples of grade JP-8 fuel (MIL-T-83133) or grade JP-5 fuel (MIL-PRF-5624), respectively, which contain the maximum allowable concentration of each inhibitor currently on the Qualified Products List for this specification. The grade JP-8 fuel (MIL-T-83133) or grade JP-5 fuel (MIL-PRF-5624) used shall contain the maximum allowable amount of fuel system icing inhibitor that conforms to MIL-I-85470. At the end of a 24-hour period, the samples shall be visually inspected for precipitation, cloudiness, or other evidence of noncompatibility.

4.4.2.2 Static dissipater additive. Grade JP-8 fuel (MIL-T-83133) filtered through clay as described in appendix X.1 of ASTM D3948 shall be blended with each static dissipater additive approved in MIL-T-83133 to provide test fuels which have a conductivity of 400 picosiemens per meter (pS/m) ±100 pS/m. After a 24-hour period, to ensure equilibrium fuel conductivity has been established, the corrosion inhibitor additive under test shall be added and mixed. At the end of another 24-hour period, no more than a ±40 percent change in the electrical conductivity of the fuel shall have occurred as a result of the test inhibitor. The fuel electrical conductivity shall be measured using either ASTM D2624 or ASTM D4308 test methods. The post-test visual inspection of the sample shall reveal no precipitation, cloudiness, or other evidence of noncompatibility. (NOTE: Some loss in fuel conductivity may occur over time when bare glass bottles or bare metal cans are used with fuels which contain static dissipator additives. The use of an epoxy-coated container is suggested. Also, fuel conductivity is temperature sensitive; no significant change in temperature should be allowed during the test.)

4.4.2.3 Thermal stability improver additive. Grade JP-8 fuel (MIL-T-83133) filtered through clay as described in appendix X.1 of ASTM D3948 shall be blended with each thermal stability improver additive approved in MIL-T-83133 at the specified concentration. The corrosion inhibitor additive under test shall be added at its maximum allowable concentration and mixed. At the end of a 24-hour period, the samples shall be visually inspected for precipitation, cloudiness, or other evidence of noncompatibility.

4.4.3 Relative effective concentration. The relative effective concentration of the inhibitor shall be determined by testing the inhibitor at various concentrations in depolarized *iso*-octane in accordance with 4.4.3.1. The inhibitor shall be tested at concentrations divisible by 3 (i.e.; 6, 9, 12, 15, ... 36 g/m^3). No intermediate concentrations shall be tested. The relative effective concentration shall be defined as the lowest concentration that provides a passing result in accordance with 4.4.3.1.6.

4.4.3.1 Rusting test method

4.4.3.1.1 Test apparatus. The test apparatus shall conform to the following requirements:

a. Oil bath that conforms to ASTM D665, with the additional requirement that it must be able to maintain the test sample at a temperature of 38.0°C ±0.5°C

b. Beaker, beaker cover, stirrer, stirring apparatus, and chuck and motors to hold and rotate specimens while polished shall conform to ASTM D665

c. Infrared heat lamp, 250 watts

d. Hypodermic syringe, glass, 3-mL, with 15.24-cm (6-inch) stainless steel needles

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e. Disposable microliter pipettes which consist of calibrated capillary tubes which contain 1, 2, 3, 4, 5, 10,15, and 20 microliters

f. Column, chromatographic, glass, 40 mm ID x 600 mm with nonmetallic stopcock. A separatory funnel, Squibb, 1-liter, with nonmetallic stopcock may be substituted for the chromatographic column.

g. Specimen holder, nonmetallic, dimensions as specified in ASTM D665 for type 2 holder

h. Specimen, dimensions as specified in ASTM D665, made of grade 1018 steel in accordance with ASTM A108. The specimen shall be fabricated from 1.58 cm (0.625 inch) diameter round stock by machining or grinding to the final diameter of 1.27 cm (0.50 inch). The specimen may be reused from test to test but shall be discarded when the diameter is reduced to 0.953 cm (0.375 inch) or less.

4.4.3.1.2 Test materials. Test materials shall conform to the following requirements:

a. Silica gel, 28-200 mesh, heated to 107.2°C for 2 hours and cooled in a dessicator before use

b. Test solvent, *iso*-octane that conforms to TT-S-735, which has been freshly depolarized as follows:

A glass chromatographic column or 1-liter separatory (Squibb) funnel is filled with silica gel to a height 20 cm above the stopcock; the silica gel is retained by a glass wool plug. (NOTE: Do not use stopcock grease.) 3.79 liters (1 gallon) of *iso*-octane is passed through the silica-gel bed by gravity, the first 50 mL discarded, and the remainder collected in a chemically-clean glass container. The depolarized *iso*-octane should be used within 1 week after treatment.

c. Test water, type B medium hard, prepared as follows: Prepare 3 stock solutions using ACS reagent-grade chemicals in distilled water. Each solution shall contain one of these chemicals: 16.4 g/L $NaHCO_3$,13.2 g/L $CaCl_2 \cdot 2H_2O$, or 8.2 g/L $MgSO_4 \cdot 7H_2O$. Pipette 10 mL of the $NaHCO_3$ stock solution into 800 mL of distilled water in a 1-liter volumetric flask, and shake vigorously. While swirling the contents of the flask, pipette 10 mL of the $CaCl_2$ stock solution and then 10 mL of the $MgSO_4$ stock solution into the flask, add distilled water to bring the volume to 1 liter, and mix thoroughly. The final blend shall be clear and free of precipitation.

d. Isopropanol, ACS reagent grade

e. Glassware cleaning solution

f. Lintless ware cleaning solution (NOTE: Cel-Fibe 1710 Wipes®, available from Cel-Fibe, Milltown, NJ, are satisfactory.)

g. Abrasive cloth, 150-, 240-, and 400-grit metal-working aluminum oxide abrasive cloth, closed coat on jeans backing. The abrasive cloth is available in rolls of 1-inch (2.54-cm) tape, the most convenient form for use in this test.

h. Disposable vinyl gloves.

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4.4.3.1.3 Specimen preparation. The specimen, whether new or reused from a previous test, shall be cleaned by solvent rinsing or brushing as needed to remove oily residues, loose rust, or foreign material. After this preliminary cleaning, the specimen shall be handled only with vinyl gloves. (NOTE: It is essential to avoid contamination of the specimen, particularly by perspiration residues. Care should be taken to avoid transfer of such contaminants to the specimen via the abrasive cloth or the lintless paper tissues.) The specimen shall then be ground successively with 150-, 240-, and 400-grit abrasive cloth while mounted in the chuck of the grinding and polishing apparatus and turned at 1700 to 1800 rpm, in accordance with the following procedures:

a. Grind with 150-grit cloth to remove all defects, irregularities, pits, and scratches, as determined by visual inspection. Old 150-grit cloth may be used to remove rust or major irregularities, but grinding shall be completed with new cloth. Stop the motor and scratch the static specimen longitudinally with 1 pass of new 150-grit cloth; use light pressure so visible scratches appear.

b. Grind with 240-grit cloth, remove all marks from the 150-grit cloth, and finish with new 240-grit cloth. Stop the motor and scratch the static specimen longitudinally with 1 pass of new 240-grit cloth; use light pressure so visible scratches appear.

c. Polish with 400-grit cloth by wrapping a strip of cloth halfway around the specimen and applying a firm but gentle downward pull to the ends of the strip. Move the strip slowly along the specimen. Shift the position of the abrasive cloth frequently to expose fresh abrasive to the specimen. Continue this procedure using new strips of abrasive cloth, as required, until all marks from the previous 240-grit operation have been removed and the surface presents a uniform appearance, free of longitudinal or spiral scratches, where all polishing marks appear to be circumferential. The final passes along the specimen shall be made with fresh abrasive cloth.

d. Remove the specimen from the chuck, wipe with lintless tissue, and store in a beaker of depolarized *iso*-octane in a desiccator that contains silica gel or other noncorrosive desiccant until ready for use. The storage period in the *iso*-octane shall not exceed 7 days.

4.4.3.1.4 Preparation of test blend. The test blend shall be prepared in the test beaker, not more than 2 hours before immersion of the specimen in the test blend. The test blend shall be prepared in accordance with the following procedure:

a. Clean the test beaker with a suitable cleaning solution (see note, below). Clean the stainless steel stirrer and methyl methacrylate beaker cover by rinsing in any suitable aliphatic hydrocarbon solvent such as a light naptha or *iso*-octane, washing thoroughly with hot distilled water, and oven drying (not over 65.6°C for cover). (NOTE: If a glass stirrer or beaker cover is used, it should be cleaned in the same manner as the test beaker. Any suitable cleaning method that provides cleaning quality comparable to the use of chromic acid may be used. The use of a detergent cleaning solution is suggested. Use stainless steel forceps to handle the glassware. Wash with tap water and then with distilled water. Rinse with reagent grade isopropyl alcohol and air- or oven-dry. Detergent cleaning avoids the potential hazards and inconvenience associated with handling corrosive chromic acid solution. The latter remains as the reference cleaning practice and as such may function as an alternative to the preferred use of detergent solutions.)

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b. Prepare the blend of *iso*-octane and inhibitor in the test beaker with direct addition of the inhibitor. No intermediate blends, concentrates, or stock solutions are permitted. Prepare each test blend using between 300 and 400 mL of *iso*-octane in the test beaker. Use pipette or pipettes to add integral numbers of microliters of the inhibitor to the beaker to increase measurement accuracy. Add the calculated volume of depolarized *iso*-octane to the test beaker. Fill the appropriate microliter pipette or pipettes with inhibitor, wipe off excess, and force the inhibitor into the *iso*-octane. Allow the pipette to fill with *iso*-octane by capillary attraction and force this rinse into the test beaker. Repeat the rinse 4 times. Calculate the amount of *iso*-octane and inhibitor to be added to the test beaker using the instructions given in 4.4.3.1.4.c.

c. Calculate the volume of *iso*-octane required for each concentration desired using the following equation, where density is in g/mL at 15°C:

$$\text{mL of } iso\text{-octane} = \frac{(\text{density of inhibitor}) \times (\text{microliters of inhibitor}) \times (1000)}{(\text{desired inhibitor concentration, grams/cubic meter})}$$

For example, assume the inhibitor has a density of 0.95 and the desired concentration is 6 grams/cubic meter of fuel. Calculate the volume of *iso*-octane required when using 2 microliters of inhibitor:

$$\text{mL of } iso\text{-octane} = \frac{(0.95) \times (2) \times (1000)}{6} = 316.7 \text{ mL}$$

For inhibitor less dense than 0.9 g/mL, the volume of *iso*-octane for many concentrations of interest will be less than 300 mL or more than 400 mL. Use the following procedure:

(1) Calculate the volume of inhibitor required for 300 mL of *iso*-octane.

(2) Increase the volume of inhibitor to the next integral microliter and add to 300 mL of *iso*-octane in the test beaker. Mix well.

(3) Calculate the amount of inhibitor/*iso*-octane blend to be removed from the test beaker to leave the desired amount of inhibitor.

(4) Remove the calculated amount of inhibitor/*iso*-octane blend and replace with an equivalent volume of depolarized *iso*-octane. Mix well. This approach is illustrated for the preceding example:

(a) Using this equation, calculate the desired volume of inhibitor for 300 mL of *iso*-octane for an inhibitor with a density of 0.85 and for a desired concentration of 6 g/m^3.

$$300 \text{ mL of } iso\text{-octane} = \frac{(0.85) \times (Z \text{ microliters of inhibitor}) \times (1000)}{6}$$

Z = 2.12 microliters of inhibitor

(b) Add the next integral volume of inhibitor (i.e., 3 microliters) to 300 mL of *iso*-octane and mix well. This gives an inhibitor concentration of 3 microliters inhibitor/300 mL *iso*-octane or 1 microliter/100 mL.

(c) The desired amount of inhibitor is 2.12 microliters. Thus, (2.12) X (100) = 212 mL of inhibitor/*iso*-octane blend is needed.

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(d) Remove 88 mL of the inhibitor/*iso*-octane blend (i.e., 300 - 212 = 88 mL). Replace with 88 mL of the depolarized *iso*-octane. Mix well. This results in the correct volume of inhibitor (i.e., 2.12 microliters) in 300 mL of *iso*-octane.

d. Place the beaker in the oil bath which has been regulated previously to maintain a sample temperature of 38.0°C ±0.5°C. The beaker is inserted in a hole of the bath cover and suspended at a level such that the oil level in the bath is not below the sample level in the beaker. Cover the beaker with the beaker cover and the stirrer in position. Adjust the stirrer so the shaft is 6 mm off-center in the beaker, and the blade is within 2 mm of the bottom of the beaker. Then suspend a thermometer through the hole in the cover intended for that purpose, so that it is immersed to a depth of 57 mm. Stir for at least 5 minutes. Turn off the stirrer. Use a clean pipette or syringe to withdraw enough test blend to leave exactly 300 mL in the beaker. Allow the test blend to come to 38.0°C ±0.5°C. Replace the thermometer with a cork or plastic plug.

4.4.3.1.5 Test procedure. After preparing a test specimen as described in 4.4.3.1.3 and a test blend as described in 4.4.3.1.4, the test shall be performed in accordance with the following procedure:

a. Remove a test specimen from the *iso*-octane in the desiccator and wipe dry with a lintless paper tissue; handle the specimen with vinyl gloves throughout this step and the following operations. Repolish with 400-grit abrasive cloth by wrapping a strip of the cloth halfway around the specimen and applying a firm but gentle downward pull to the ends of the strip. Move the strip slowly along the specimen, twice in each direction, shifting the strip after the first back-and-forth pass so fresh abrasive is exposed to the specimen. Inspect the specimen to ensure the surface presents a uniform appearance, free of longitudinal or spiral scratches, where all polishing marks appear to be circumferential. Additional polishing is required if the specimen appearance is other than described. After polishing is completed, remove the specimen from the chuck, wipe lightly with lintless paper tissue, and screw the specimen into the specimen holder. Rinse the specimen with a stream of isopropanol from a wash bottle. Wipe dry immediately, wiping twice with fresh lintless paper tissues, using firm pressure and rotating the specimen while drawing through the tissue. (NOTE: Under conditions of high ambient humidity, it is necessary to heat the specimen to prevent condensation of moisture and premature rusting. Under such conditions, place the specimen and holder 6 inches from a 250-watt infrared heat lamp and rotate for 1 minute before the rinsing operation. Keep the specimen under the lamp while proceeding with the rinsing and wiping operations.)

b. Immediately after rinsing and wiping, insert the specimen and holder through the specimen hole in the beaker cover and suspend the specimen so that its lower end is 13 to 15 mm from the bottom of the beaker. Leave the specimen in the test blend for a 10-minute static soak, then turn on the stirrer and soak dynamically for 20 minutes. (NOTE: When multiple tests are run simultaneously, it is permissible to extend the static soak period to not more than 40 minutes in the case of the “first-in” specimen, giving the “last-in” specimen a 10-minute soak.)

c. Turn off the stirrer. Remove the cork or plug from the beaker cover. Carefully add 30 mL of test water to the bottom of the test beaker by means of a hypodermic syringe. Change to a clean needle for each test beaker. Replace the cork or plug in the beaker cover.

d. Start the stirrer immediately and run for 5 hours; hold the bath temperature at the same setting so the test samples will be maintained at 38.0°C ±0.5°C.

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e. At the end of 5 hours, stop the stirrer, remove the specimen and holder, rinse immediately with isopropanol, and allow to air-dry. Examine at once without magnification under normal indoor illumination, approximately 646 lux or 60 footcandles. Scan the surface very carefully to detect any small pits. Record observations of visible rust, pits, stains, or deposits.

4.4.3.1.6 Interpretation of test results. A test shall be reported as failed if the center 48-mm section of the specimen shows 6 or more rust spots of any size, or if it shows any rust spot 1 mm in diameter or larger. (NOTE: The ends of the specimen, outside the center section, are ignored in rating the specimen.) Visible deposits or stains other than rust shall not constitute failure; deposits or stains may be examined microscopically to determine their classification. In order to assign a pass-fail rating to a given inhibitor at a given concentration, two tests shall be performed. The inhibitor shall be reported as passing at the given concentration if both tests give passing ratings, or failure at the given concentration if both tests give failing ratings. If the two tests give one passing rating and one failing, two additional tests shall be performed. If either or both of these additional tests give a failing rating, the inhibitor shall be reported as failing at the given concentration. If both of the additional tests give passing ratings, the inhibitor shall be reported as passing the given concentration.

4.4.4 Minimum effective concentration

4.4.4.1 Lubricity test. The inhibitor shall be tested in accordance with the BOCLE as described in ASTM D5001, using Isopar M as the solvent and a 1000-gram load on the ball.

4.4.5 Maximum allowable concentration

4.4.5.1 Micro-Separometer Rating. The inhibitor shall be blended into the reference fluid base as described in ASTM D3948 and tested in accordance with ASTM D3948. For any given concentration of inhibitor, the average of three test results shall be used to determine the conformance to the requirements of 3.7 or 4.3.

4.4.6 Ash content determinations. The ash content of the inhibitor shall be determined in accordance with ASTM D482, using a platinum or porcelain crucible.

4.4.7 Pour point determination. Pour point shall be determined in accordance with ASTM D97, ASTM D5949, ASTM D5950, or ASTM D5985.

4.4.8 Aircraft engine test. The engine shall be operated for 100 hours in accordance with the engine operating requirements of MIL-PRF-7808. The fuel, grade JP-8 fuel that conforms to MIL-T-83133 or grade JP-5 fuel that conforms to MIL-PRF-5624, shall contain twice the maximum allowable concentration of the inhibitor. Upon completion of the test, components of the engine exposed to the fuel—such as fuel controls, fuel nozzles, combustion section, turbine blades, exhaust section, elastomers, fuel/oil heat exchangers, and fuel pumps—shall be examined for evidence of excess wear, deposits, corrosion, or other deleterious effects. This test shall be performed by the activity responsible for qualification (see 6.6).

4.4.9 Specification tests. The inhibitor shall be added at its maximum allowable concentration to a base fuel that contains no inhibitor but is otherwise representative of each grade of fuel for which the additive is to be qualified. The blend of fuel and inhibitor shall be subjected to all of the tests of each applicable specification.

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4.4.10 Storage stability test. Two 1-liter or 1-quart amber glass bottles shall each be filled with 850 mL of the inhibitor and shall be tightly capped by means of a screw cap that has a conical, polyethylene liner. Each bottle shall be wrapped in a minimum amount of opaque packing materials sufficient for protection against mechanical damage, but minimal in thermal insulation qualities. The wrapped bottles shall be enclosed in a tight wooden or metal box for further protection against breakage and sunlight. The crated samples shall be stored at ambient outdoor conditions in a temperate climate. The box shall be kept off the ground and protected from direct sunlight and precipitation under a canopy, open shed roof, or similar ventilated shelter. The crated samples shall be stored undisturbed in an upright position for the specified period. One of the samples shall be stored for exactly 12 months and then removed for examination and testing; the other sample shall be stored for 12 months or less and may be removed for examination and testing at any time, at the option of the Qualifying Activity. Whenever a sample is removed for examination and testing, it shall be uncrated with minimum disturbance; the bottle shall not be shaken, inverted, or otherwise agitated. The contents of the bottle shall be inspected visually for precipitation, separation into layers, or other evidence of gross separation. The presence or absence and the nature of such separation shall be recorded. The top half of the liquid sample shall be carefully removed by suction or siphoning into another bottle, without disturbing the bottom half of the original sample. The top-half sample, after transfer to the second bottle, shall be shaken thoroughly and then used in laboratory testing performed in accordance with 3.12. The bottom-half sample, in the original storage bottle, shall be retained for examination and possible additional testing to detect changes caused by storage.

4.4.11 Induction system deposit test. The inhibitor, at its maximum concentration, shall be blended into a MIL-G-3056 motor gasoline. The test fuel shall then be tested for the formation of induction system deposits in accordance with Method 500 of FED-STD-791. The naphtha-washed deposits shall not exceed 2 mg/100 mg of fuel. The MIL-G-3056 gasoline without inhibitor shall also be tested in accordance with Method 500 of FED-STD-791 concurrently to define the level of deposition occurring as a result of the inhibitor.

4.4.12 Emulsification tendency test. The inhibitor, at maximum allowable concentration, shall be blended into a MIL-G-3056 motor gasoline and a MIL-F-46162 diesel fuel. Each test fuel shall then be examined for emulsification tendencies in accordance with Method 550 of FED-STD-791. Interface ratings in excess of 3 are evidence of unsatisfactory emulsification tendencies and shall not be allowed. The MIL-G-3056 motor gasoline and the MIL-F-46162 diesel fuel shall also be tested in accordance with Method 550 of FED-STD-791 to identify the quality of the fuels before the addition of the inhibitor.

4.4.13 Accelerated stability test. The test inhibitor, at its maximum allowable concentration, shall be blended into a diesel fuel (A-A-52557) that contains no additives. Each test fuel shall be tested for the formation of total insolubles in accordance with ASTM D2274. The total insolubles shall not exceed 1.5 mg/100 mL. The diesel fuel without the test inhibitor shall also be tested in accordance with ASTM D2274 concurrently to define the level of insolubles occurring without the presence of the inhibitor. (NOTE: A suitable reference diesel fuel for this evaluation is described in Method 341.4 of FED-STD-791.)

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4.4.14 Identification tests. Identification tests shall be conducted in accordance with the following methods:

Density at 15°C	ASTM D1298 or pycnometer or ASTM D4052
Viscosity at 40°C	ASTM D445
Flash Point	ASTM D56 or ASTM D93
Neutralization Number	ASTM D664, total acid number
pH	On a 0.10 - 0.11 gram sample in 125 mL of ASTM D664 titration solvent. Read the constant pH as defined in note 15 of ASTM D664.
Metallic constituent	Emission spectrograph, not applicable for materials with ash contents of 0.05 percent or lower.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The inhibitors covered by this specification are used, when specifically authorized, in jet engine fuels for the prevention of corrosion in fuel handling, transportation, and storage equipment and to improve the lubricating qualities of jet fuels. Certain of the inhibitors are also used in automotive gasoline, diesel fuel, and related petroleum products.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. Quantity required.
d. Packaging requirements (see 5. 1).

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6.2.1 Amount of inhibitor use. When Government procurement documents specify the use of inhibitors in fuels and related petroleum products, the concentration of inhibitor will be specified in grams of inhibitor per cubic meter of fuel and will not be less than the minimum effective concentration nor more than the maximum allowable concentration as listed on the Qualified Products List. Since the inhibitor is intended for use under many different environments, it is not possible to establish a single, optimum concentration for all uses. Therefore, when a specific concentration is not required by the Government, the quantity of inhibitor used may vary to meet specific conditions.

6.3 Material Safety Data Sheets. Contracting officers will identify those activities which require copies of completed Material Safety Data Sheets (MSDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.4 Toxicity. Questions pertinent to toxicity should be referred by the Procuring Activity to the appropriate department medical service, which will act as an advisor to the Procuring Activity. The U.S. Army Center for Health Promotion and Preventive Medicine (USACHPPM) will act as advisor to Army procuring activities. Army toxicity questions should be addressed through: Commander, Headquarters, U.S. Army Materiel Command, ATTN: AMCSG-I, 5001 Eisenhower Avenue, Alexandria, VA 22333-0001 for Commander, U.S. Army Center for Health Promotion and Preventive Medicine, ATTN: MCHB-DC-TTE, 5158 Blackhawk Road, Aberdeen Proving Ground MD 21010-5422.

6.5 Inhibitor for addition to fuels. When a fuel contractor or the Government purchases the inhibitor for addition to fuels to be used by the Government, the manufacturer of the inhibitor must certify to the purchaser that the product is an inhibitor that has been qualified under this specification. In addition, a test report that shows compliance of the product with the requirements of 4.3 must be supplied to the purchaser. Additional data may be required by the Purchasing Activity to establish compliance with this specification.

6.6 Qualification. With respect to products which require qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL) No. 25017 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Wright Laboratory, WL/POSF, Bldg. 490, 1790 Loop Road North, Wright-Patterson AFB OH 45433-7103.

6.6.1 Qualification test report. A certified test report from the independent laboratory will be requested by the Qualifying Activity. The test report will contain laboratory data which detail the results required by 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.11, 3.12, and 3.16. The test report will also contain laboratory data on any of the special tests conducted to qualify the inhibitor for use in motor gasolines and diesel fuels (i.e.; 3.13, 3.14, and 3.15). In addition, complete formulation data will be supplied to the Qualifying Activity. This data will include chemical composition (I.U.P.A.C. nomenclature and structural diagrams of each ingredient), the percentages of each ingredient, the manufacturer and trade names of each ingredient, and, where available, the purity of each ingredient. The contractor will be required to furnish toxicological data and formulations necessary to evaluate the safety of the material for the proposed use.

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6.7 Conversion of metric units. Units of measure have been converted to the International System of Units (Metric) in accordance with ASTM E380. If test results are obtained in units other than Metric or there is a requirement to report dual units, ASTM E380 or ASTM D1250 Volume XI/XII should be used to convert the units.

6.8 Subject term (key word) listing.

Diesel
Fuel additive
Fuel lubricity additive
Gasoline
Isopropanol
Petroleum
Turbine fuel

6.9 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement STANAG 3390. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the Preparing Activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AT
Navy – AS
Air Force – 11

Review activities:
Army – AV, MD1
Navy – SH
Air Force – 68
DLA – GS, PS

Preparing activity:
Air Force – 11
(Project 6850-1204)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 45 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, not to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-25017F	2. DOCUMENT DATE (YYMMDD) 971110

3. DOCUMENT TITLE
Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

A. NAME Air Force Code 11 Technical Information & Documents Section	d. TELEPHONE (Include Area Code) (1) Commercial (937) 255-6295 (2) DSN (If applicable) 785-6295
C. ADDRESS (Include Zip Code) ASC/ENSI 2530 Loop Road West Wright-Patterson AFB OH 45433-7101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD FORM 1426, OCT 89 PREVIOUS EDITIONS ARE OBSOLETE.

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